

Keeping it natural

Natural materials, such as rubber, cotton, and different types of wood, are used in a wide range of everyday items, such as the ones seen here.

Vulcanized tires

Adding sulfur to natural rubber, a process called vulcanization, increases its durability.



Elastic, not plastic

Rubber gloves are often made of flexible latex.

Thin but strong

Cellulose polymer chains line up together to give cotton thread its strength.



Absorbent cotton

Cotton is great for towels and cotton swabs as it is soft and can absorb up to 27 times its weight in water.

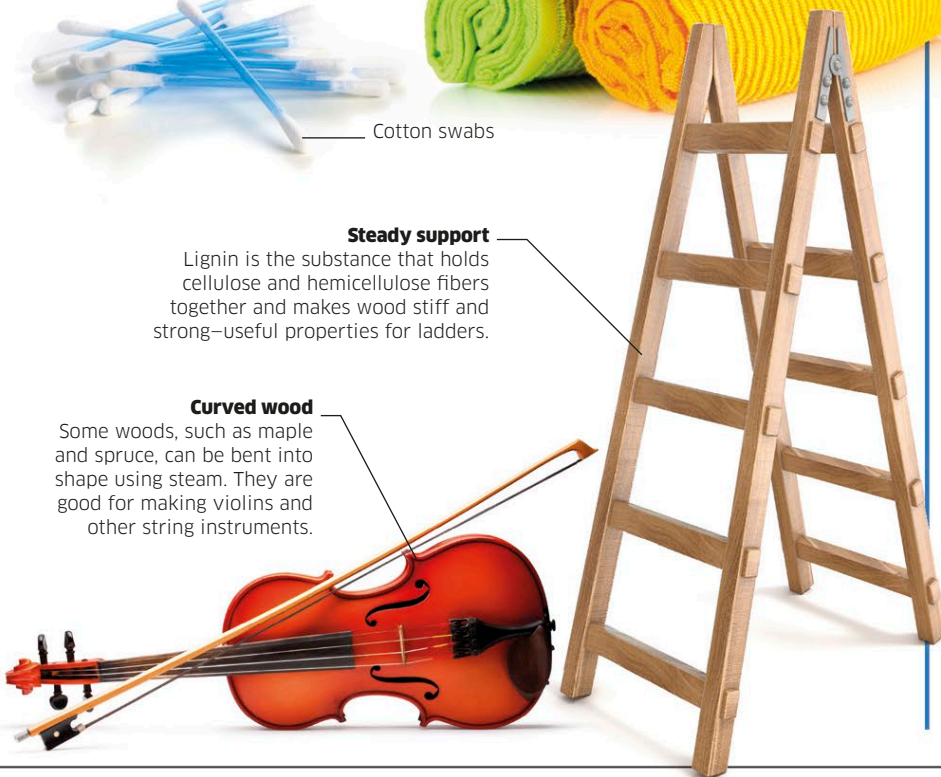


Cotton swabs



Steady support

Lignin is the substance that holds cellulose and hemicellulose fibers together and makes wood stiff and strong—useful properties for ladders.



Curved wood

Some woods, such as maple and spruce, can be bent into shape using steam. They are good for making violins and other string instruments.

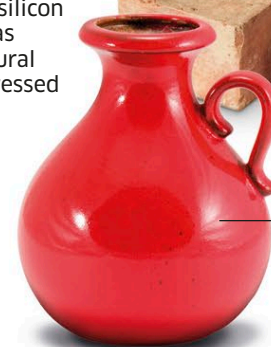
Materials from Earth's crust

Earth materials range from sand, clay, and rocks to minerals and metals. Materials from the earth have always been important for building. If you look at buildings, you can usually see what materials lie underground in the area—flint or slate, sandstone, limestone, marble, or clay. These materials are also essential for practical and decorative cookware, earthenware, and utensils.

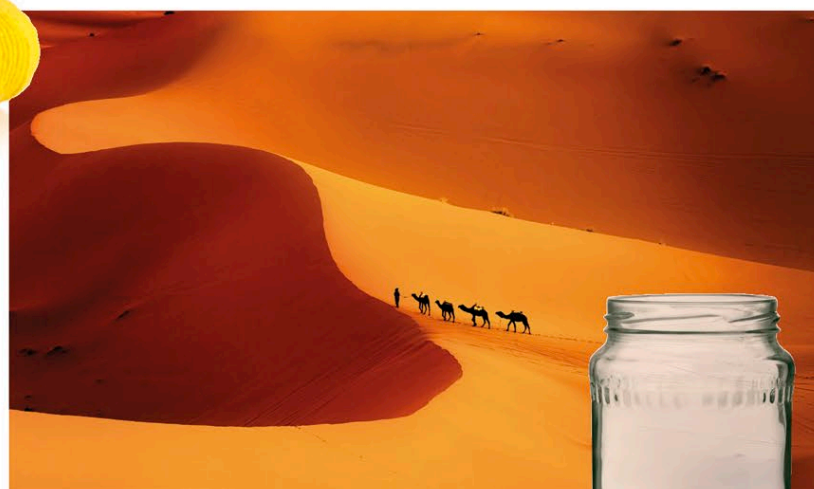


Clay and clay products

Clay, a mixture of the minerals silicon dioxide and aluminum oxide, has many uses. To make bricks, natural clay is mixed with water and pressed into shape before being dried. It is then baked at very hot temperatures to make it waterproof. Pottery is made in a similar way, but with clay of finer particles.



Earthenware pottery is fired at temperatures of around 1,830°F (1,000°C).



Sand and glass

Glass is made from sand. It is usually the sand common in deserts, which consists of the mineral silica. Beach sand often has traces of other substances, making less clear glass. Carefully chosen additives color the glass. The ingredients are melted together at 2,732°F (1,500°C) before being shaped into window panes, drinking glasses, or bottles.



Eyeglass lenses used to be made of pure glass. Today they are often plastic.